Response to Supervisor Comments

Manuscript Title: *Mitigating Physical Layer Vulnerabilities in 6G Communications using Deep Reinforcement Learning*

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Comment 1: Introduce Additional Baselines

Comment

Currently, the DQN is only compared to a static baseline, showing an average 49.28% improvement. To make the evaluation more compelling, the framework should be compared with an established non-AI heuristic algorithm or a simpler baseline reinforcement learning method.

Response

We thank you for this valuable suggestion. We agree that evaluating the proposed framework against only a single baseline may not sufficiently demonstrate its effectiveness across different decision-making strategies.

Revision Made

A second baseline, namely the **Threshold-Based Heuristic Policy (TBHP)**, was implemented and incorporated into the experimental evaluation. The TBHP applies domain-specific rules by increasing transmission power when signal strength is low, injecting artificial noise when the eavesdropper is nearby, and adjusting beamforming otherwise. The proposed DQN framework was evaluated against both the static random policy and TBHP. Experimental results show that the DQN achieved an average improvement of **35.84%** over TBHP while maintaining an average improvement of **49.28%** over the static baseline.

Comment 2: Clarify Simulation Parameters

Comment

Since the evaluation uses a custom lightweight Python simulation, a detailed description of the simulation environment and parameters is required. This should include communication distances and relevant environment settings to improve reproducibility.

Response

We appreciate this important recommendation. Reproducibility is a key aspect of experimental research, and additional implementation details help readers better understand and validate the proposed framework.

Revision Made

A new table entitled **“Simulation Environment and Training Hyperparameters”** was added to the manuscript. The table provides a complete summary of the simulation and training setup, including state-space dimensionality, variable ranges, communication distances, jammer and noise dynamics, neural network architecture, optimizer settings, learning rate, replay buffer size, exploration strategy, training timesteps, and evaluation runs.

Comment 3: Expand on the State Space Representation

Comment

While the neural network architecture is described, the dimensions and numerical representation of the state-space variables are not fully defined. Additional clarification is required regarding how the state variables are vectorized and provided to the model.

Response

Thank you for highlighting this issue. We agree that a more detailed description of the state representation improves the clarity and reproducibility of the proposed methodology.

Revision Made

A dedicated subsection entitled **"State Space Vectorization"** was added. The manuscript now explicitly defines the state vector as:

$$S_t = [s, p_j, n, d_e]$$

where:

- (s) represents normalized signal strength,
- (pi) represents jammer interference power,
- (n) represents normalized noise level,
- (de) represents the normalized distance between the eavesdropper and the legitimate user.

The revised manuscript further explains how these values are represented as a normalized float32 vector and directly provided as input to the DQN neural network.

Comment 4: Fix Formatting Anomalies in Data

Comment

Several formatting inconsistencies exist in the results table. Numerical values should be standardized using decimal points and formatted according to IEEE guidelines.

Response

We appreciate this careful observation and thank you for identifying these formatting issues.

Revision Made

All numerical values reported in the experimental results were reviewed and standardized according to IEEE formatting conventions. Decimal separators were corrected, inconsistent numeric representations were removed, and the results table was reformatted to ensure consistency and readability throughout the manuscript.

Additional Improvements Implemented

In addition to the requested revisions, several further improvements were introduced to strengthen the manuscript:

- Updated the Markov Decision Process formulation to include the missing eavesdropper-distance variable.
- Explicitly defined the reward-function coefficients $\alpha = 5$, $\beta = 1$, and $\gamma = 0.5$.
- Added a detailed explanation of the secrecy-rate calculation used in the prototype environment.
- Updated the Abstract, Contributions, Experimental Evaluation, and Conclusion sections to reflect the new two-baseline comparison.
- Regenerated Figure 2 to include DQN, Static Random Policy, and Threshold-Based Heuristic Policy (TBHP).
- Removed redundant text and unused LaTeX packages to improve manuscript consistency.
- Improved the overall clarity, readability, and organization of the manuscript.

We sincerely thank you for your valuable feedback and constructive suggestions. The revisions significantly improved the quality, clarity, and reproducibility of the manuscript.